

Assignment 01

1. There are around 4.3×10^{19} possible configurations of a 3 x 3 Rubik's cube. However, if played optimally, any configuration can be solved in 20 moves or less. Here, one single move consists of a rotation of one of the faces of the cube. There are 27 possible rotations from a single configuration.

- a. Suppose that you are building a robot that will solve this Rubik's cube in the fastest way possible. Write down the PEAS description for it.

- b. Formulate this as a search problem. Your formulation must contain the state representation, total possible number of states, actions, successor function, start state, and goal test.

2. Consider a Coffee Making Robot which can make a cup of coffee based on voice command from user. Now write the PEAS description for it.

Imagine you are designing an intelligent agent to plan the optimal route for a delivery drone to deliver packages in a city. Give a formal description of this problem as a search problem

3. Okabe Rintarou has a time machine that allows him to jump forward or backward in time by any number of years in a single jump. He is currently in the year 2025 and wants to reach the year 2000 using at most 4 time jumps. Each jump can move him to any year in the past or future. For example, the sequence $2025 \rightarrow 2001 \rightarrow 1997 \rightarrow 2000$ reaches the goal in 3 jumps. Now provide a problem formulation for this scenario.

4. Answer the following questions:

- a. Suppose you are trying to solve the 8 queens problem using greedy hill climbing search. How will the performance be affected if you allow sideways move (choose equal valued successors if better ones are not generated)? [2]

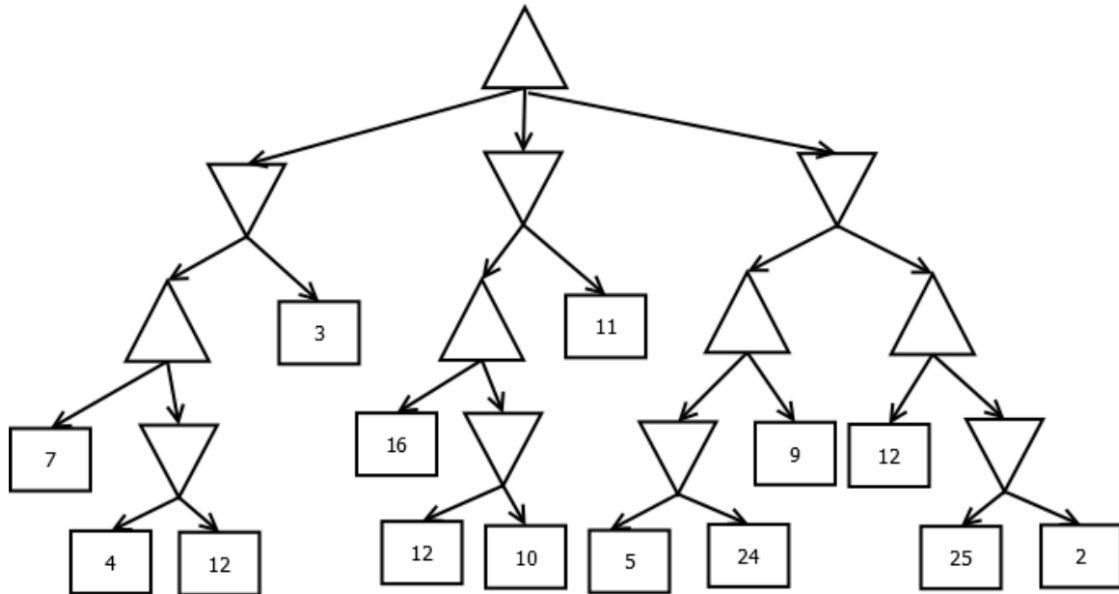
- b. What is the difference between Randomly Restarting Hill Climbing Search k times and local beam search with a beam size of k? [2]

- c. What will happen in the local beam search if we set the beam size(k) to be 1? [1]

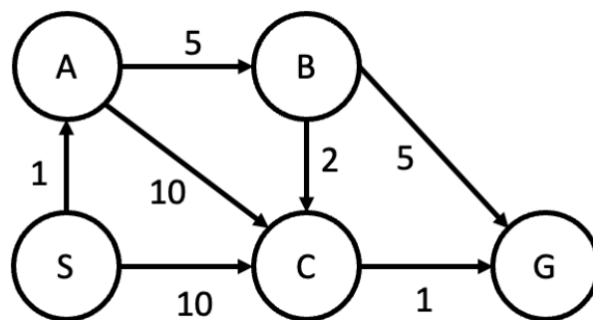
- d. What is the role of the temperature, T in Simulated Annealing? How do chances of taking bad moves change with the value of T? [2]

5.

Apply minimax search and show which nodes will be pruned if you use alpha-beta pruning.



6.



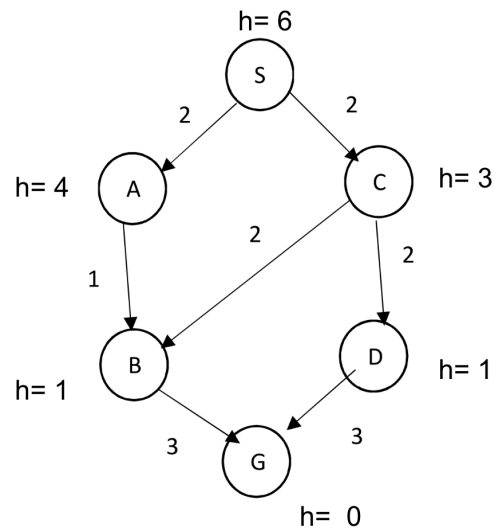
Apply BFS, DFS, UCS for this graph and out the solution path and cost where s is the start node and G is the goal node using graph search (**consider this as an undirected graph**)

7. Show the solution paths and the associated actual path costs for the following search algorithms:

i. A* Tree Search

ii. A* Graph Search

iii. Greedy Best First Search



- Is A* graph search or A* tree search guaranteed to provide the optimal solution here? Why or why not? Explain briefly.
- If both A* and UCS provide optimal solution, however, in this case, which one would you prefer among A* search and UCS? And why?
- If you have to choose one particular search among all three, which one would you choose and why?